

# Modeling 2D Marshak Waves

Optional Subtitle

Yuval Kochavi

The Hebrew University of Jerusalem

June 17, 2026

- 1 Describing the Physical system
- 2 Marshak Waves Experiments
- 3 1.5D model
  - 1D effects
  - 2D effects
- 4 Full 2D model
  - Modeling the front curvature
  - Comparing to 2D supersonic diffusion simulation
  - A full 2D Temperature distribution
- 5 Conclusion

# Photon Transport and Energy Exchange

Photon transport (Boltzmann equation):

$$\frac{1}{c} \frac{\partial I}{\partial t} + \hat{\Omega} \cdot \nabla I + (\sigma_a(T_m) + \sigma_s(T_m))I = \sigma_a(T_m)B(T_m) + \frac{\sigma_s(T_m)}{4\pi} \int_{4\pi} I d\hat{\Omega} + S(\hat{\Omega}, \vec{r}, t) \quad (1)$$

Energy exchange between radiation and matter:

$$\frac{C_v(T_m)}{c} \frac{\partial T_m}{\partial t} = \sigma_a(T_m) \left( \frac{1}{c} \int_{4\pi} I d\hat{\Omega} - aT_m^4 \right) \quad (2)$$

# Gray Coupled Equations and LTE

We get the "gray" coupled equations:

$$\begin{aligned}\frac{\partial E}{\partial t} - \nabla \cdot \left( \frac{c}{3\sigma} \nabla E \right) &= c\sigma(U - E), & U &= aT_m^4, & E &= aT_R^4 \\ \frac{\partial e}{\partial t} &= c\sigma(E - U), & e &= fT_m^\beta \rho^{-\mu}\end{aligned}$$

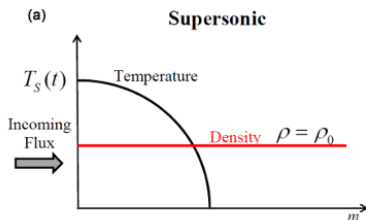
Finally, assuming complete thermodynamic equilibrium (CTE) between the radiation field and the material ( $T_R = T_m$ ):

$$\frac{\partial T^4}{\partial t} + f\rho^{-\mu} \frac{\partial T^\beta}{\partial t} = \nabla \cdot \left( \frac{c}{3\sigma} \nabla (aT^4) \right) \quad (3)$$

Assuming  $\rho f T^\beta \rho^{-\mu} \gg aT^4$  (True until around 3keV):

$$f\rho^{-\mu} \frac{\partial T^\beta}{\partial t} = \nabla \cdot \left( \frac{c}{3\sigma} \nabla (aT^4) \right)$$

# Motivation



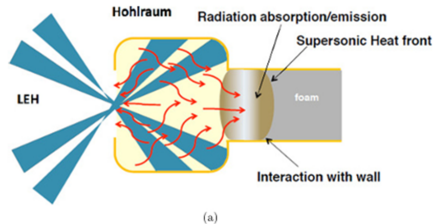
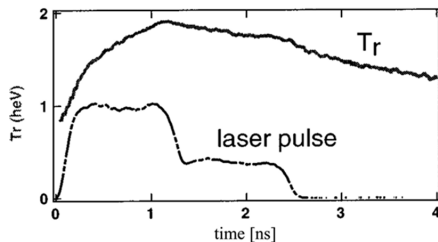
- Given the material properties equations:

$$e = fT^\beta \rho^{-\mu}, \quad \frac{1}{\kappa} = gT^\alpha \rho^{-\lambda}$$

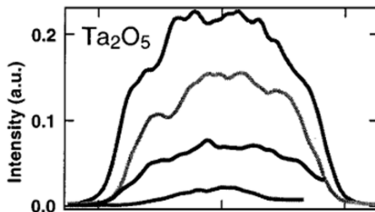
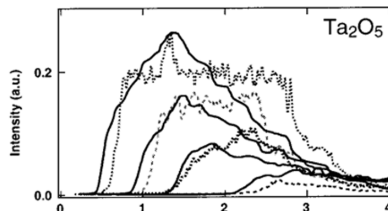
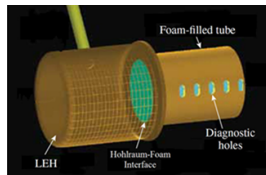
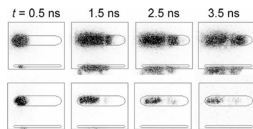
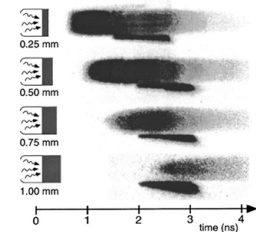
- The diffusion coefficient  $D$  is not linear.
- Because of this, the entire system becomes highly non-linear and hard to solve analytically.
- Our Motivation:** To build a robust model and fundamentally understand the dynamics of this system.

# Marshak Waves Experiments

Using a Hohlraum as an x-ray generator, a supersonic heatwave propagates in the low-density foam.



# Marshak Waves Experiments



# Experiments tested

| The experiment                 | foam type                                          | density<br>(mg/cc) | temperature<br>(eV) |
|--------------------------------|----------------------------------------------------|--------------------|---------------------|
| Massen <i>et al.</i>           | $C_{11}H_{16}Pb_{0.3852}$                          | 80                 | 100–150             |
| Back <i>et al.</i>             | $SiO_2$ , $Ta_2O_5$                                | 10–50              | 85–190              |
| Xu <i>et al.</i>               | $C_6H_{12}$ ,<br>$C_6H_{12}Cu_{0.394}$             | 50                 | 160                 |
| Ji-Yan <i>et al.</i>           | $C_8H_8$                                           | 160                | 175                 |
| Rosen and Keiter <i>et al.</i> | $C_{15}H_{20}O_6$ ,<br>$C_{15}H_{20}O_6Au_{0.172}$ | 45–70              | 200–210             |
| Moore <i>et al.</i>            | $SiO_2$ , $C_8H_7Cl$                               | 90–120             | 310                 |
| Courtois <i>et al.</i>         | $SiO_2$                                            | 19 – 29            | 170                 |



# 1.5D model

# HR Model - Governing Equations

The "Black" diffusion equation (neglecting radiation energy density  $\frac{\partial E}{\partial t} \rightarrow 0$ ):

$$\underbrace{\frac{\partial E}{\partial t}}_{\rightarrow 0} + \frac{\partial e}{\partial t} = \frac{\partial}{\partial x} \left( \frac{c}{3\rho\kappa} \frac{\partial(aT^4)}{\partial x} \right)$$

With specific energy and opacity models:

$$e = fT^\beta \rho^{-\mu}, \quad \frac{1}{\kappa} = gT^\alpha \rho^{-\lambda}$$

where  $f, \mu, g, \lambda, \alpha, \beta$  are material-dependent parameters.

HR solved the diffusion equation using perturbation theory and found some interesting results.

# HR Model - Heat Front Solution

The heat front position  $x_F(t)$  is given by:

$$x_F^2(t) = \frac{2 + \varepsilon}{1 - \varepsilon} C H^{-\varepsilon}(t) \int_0^t H(t') dt'$$

where:

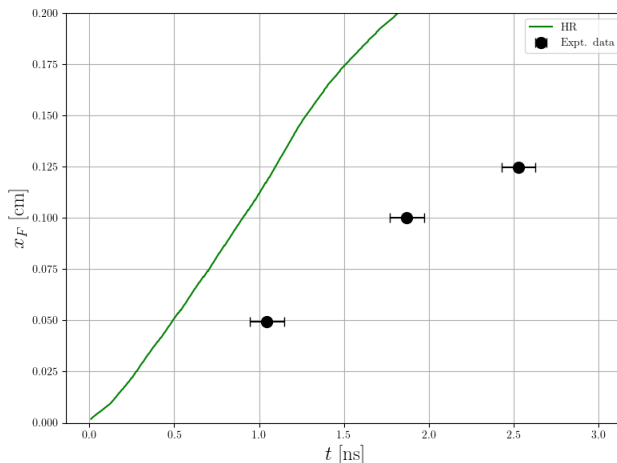
$$C = \frac{16g\sigma_{SB}}{3f(4 + \alpha)} \rho^{2-\mu-\lambda}; \quad \varepsilon = \frac{\beta}{4 + \alpha}$$

$$H(t) = T_s^{4+\alpha}(t)$$

where  $T_s$  is the surface temperature at  $x = 0$ .

# HR 1D Model Limitations

HR misses the real experimental results of Back et al. by a factor of 2!



# 1.5D model

1D effects

# Marshak Boundary Condition: $T_D \neq T_s$

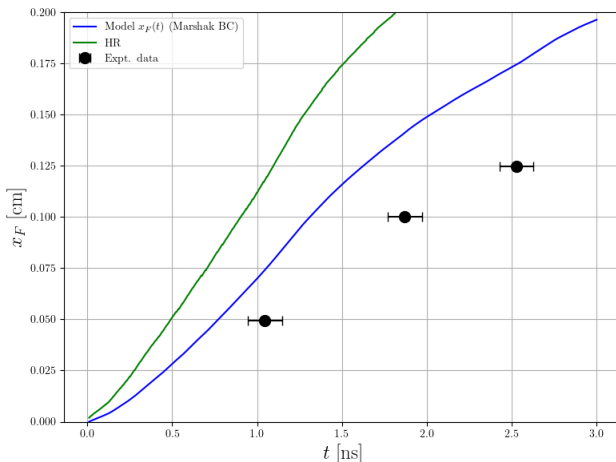
- Assuming  $T_s = T_D$  overestimates the wave speed.
- Instead, energy conservation at the boundary requires:

$$\sigma_{SB} T_D^4(t) = \sigma_{SB} T_s^4(t) + \frac{F(0, t)}{2}$$

where  $F(0, t)$  is the net flux entering the foam.

# Slowing the Front: Marshak Boundary Condition

Applying the Marshak boundary condition slows the propagation front significantly:



# 1.5D model

2D effects



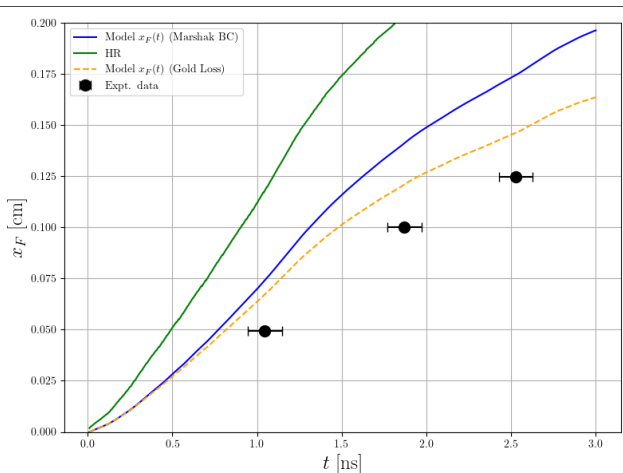
- Energy is lost from the foam into the tube walls.
- Can be modeled for different wall materials: **Gold**, **Copper**, **Beryllium**, and **Vacuum**.
- For **Gold**:

$$E_{\text{gold}}(t) = 0.59 T_s^{3.35} [t - t_0(x)]^{0.59} \quad [\text{hJ/mm}^2]$$

where  $t_0(x)$  is the front arrival time at position  $x$ .

# Effect of Wall Losses

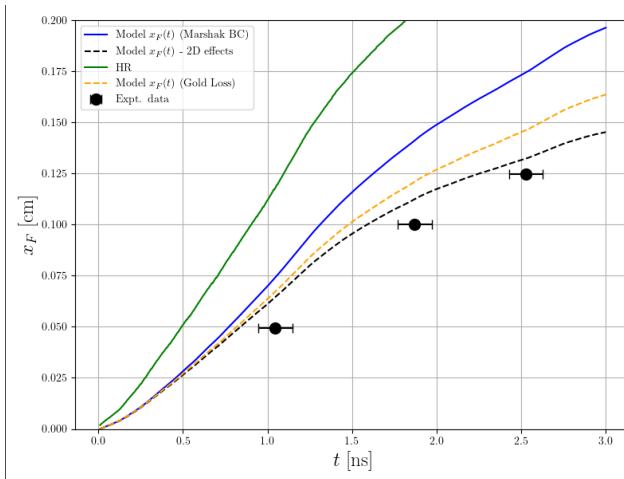
Including wall loss reduces the front velocity:



- High-Z wall material (e.g. Gold) absorbs energy and ablates inward into the foam.
- **Changing Foam Radius:** The inlet and tube radius shrink, reducing the heat wave cross-section.
- **Changing Foam Density:** The ablated wall material compresses and densifies the adjacent foam.
- Both effects must be incorporated to model the dynamic tube geometry.

# Effect of Wall Ablation

Inward wall ablation compressively densifies the foam, which significantly slows down propagation:



# Effective Mean Free Path

When  $\lambda_R \geq 2R$ , wall collisions control the transport instead of diffusion.  
Can be modeled as:

$$\frac{1}{\lambda_{\text{eff}}} = \frac{1}{\lambda_R} + \frac{1}{2R}$$

where  $\lambda_R$  is the Ross Mean Free Path.

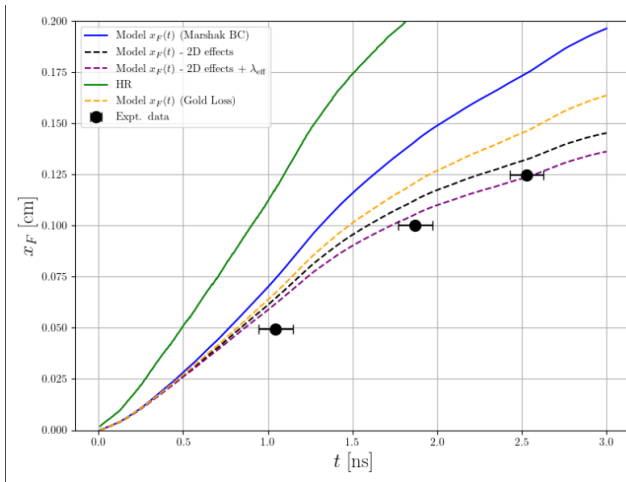
A smoother transition is defined using the generalized power mean:

$$\lambda_{\text{eff}} = ((2R)^{-n} + \lambda_R^{-n})^{-1/n}$$

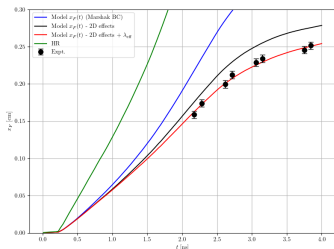
where  $n$  controls the "tightness" of the transition.

# Effective Mean Free Path Results

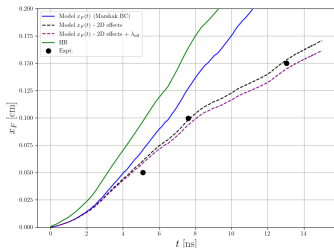
Including the effective mean free path correction matches the simulation to the experimental data at early times:



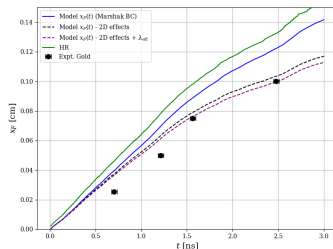
# Comparison to many experiments



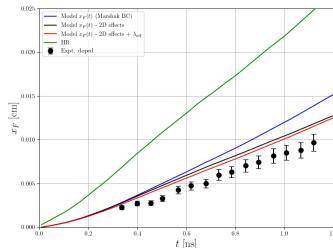
Moore et al. JQSRT, (2015)



Back et al. PRL, (2000)



Back et al. POP, (2000)



Ji-Yan et al. CPB, (2010)

# Full 2D model

Modeling the front curvature



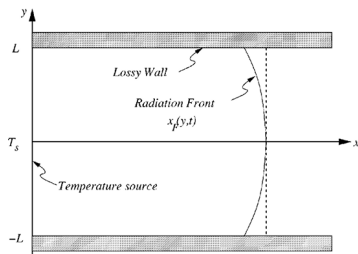
# Hurricane & Hammer: understanding the front curvature

Hurricane & Hammer (2006) solved the laplace equation (assumed constant opacity and neglected  $e$ ):

$$\underbrace{\frac{\partial e}{\partial t}}_{\rightarrow 0} = \frac{4}{3} \nabla \cdot \left( \frac{1}{\rho \kappa} \nabla (T^4) \right)$$

and got the 2D front in Cartesian coordinates (where  $\varepsilon = \frac{3}{4} \rho \kappa L (1 - a)$ ):

$$x_F(y, t) = \frac{L}{\varepsilon} \cosh^{-1} \left( 1 + \frac{D \varepsilon t}{L^2} \right) \cos \left( \frac{\sqrt{\varepsilon}}{L} y \right) + \text{HOT}$$



# Cylinder Coordinates

The model was re-developed in cylindrical coordinates (French group) to match cylindrical geometries:

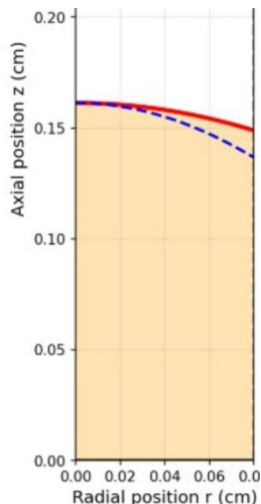
$$x_F(r, t) = \frac{1}{\kappa_0} \cosh^{-1} \left( 1 + \frac{D\kappa_0^2 t}{2} \right) J_0(\kappa_0 r) + \text{HOT}$$

**Planar  $\longleftrightarrow$  Cylindrical Analogy:**

$\cos(ky) \longleftrightarrow J_0(\kappa r)$  For small  $\varepsilon$ :

$$\kappa_0 \approx \frac{\sqrt{2\varepsilon}}{R}, \quad \varepsilon = \frac{3}{4} \rho \kappa R (1 - a)$$

where  $a$  is the wall albedo.



# Applying on our model: Dynamic Albedo

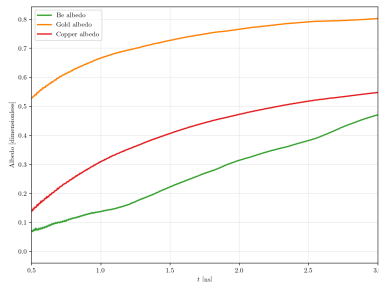
Until now, wall albedo was treated as a constant parameter. We update it dynamically in time:

$$\alpha(t) = \frac{F_{\text{out, wall}}}{F_{\text{in, wall}}} = \frac{\sigma_{SB} T_{s, \text{interface}}^4 - \frac{dE_{\text{wall}}}{dt} / 2}{\sigma_{SB} T_{s, \text{interface}}^4 + \frac{dE_{\text{wall}}}{dt} / 2}$$

This provides a much more physical description of the energy reflection.

where the curvature goes as

$J_0 \left( \frac{\sqrt{2\varepsilon(t)}}{R} \right)$ , and  $\varepsilon(t)$  scales with  $(1 - \alpha(t))$ .



**Problem: the longitudinal  
front is way off!**

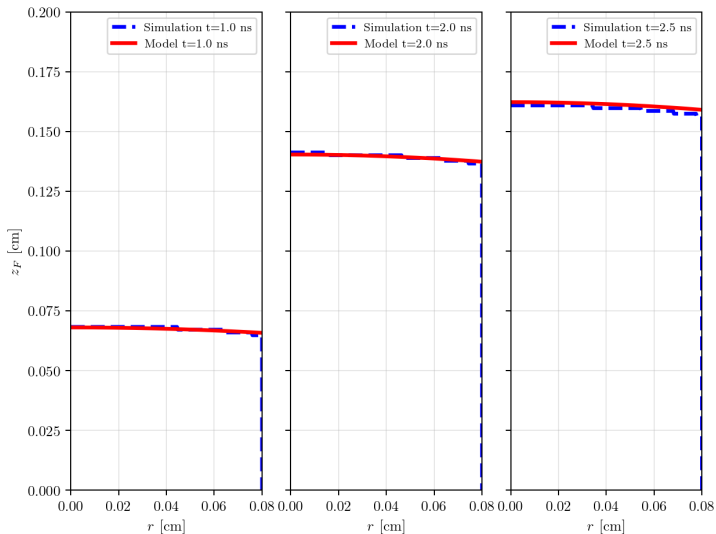
**Luckily, we have a direct  
longitudinal position:**

$$x_F(r, t) = x_{1.5D}(t) \cdot J_0(\kappa_0(t)r)$$

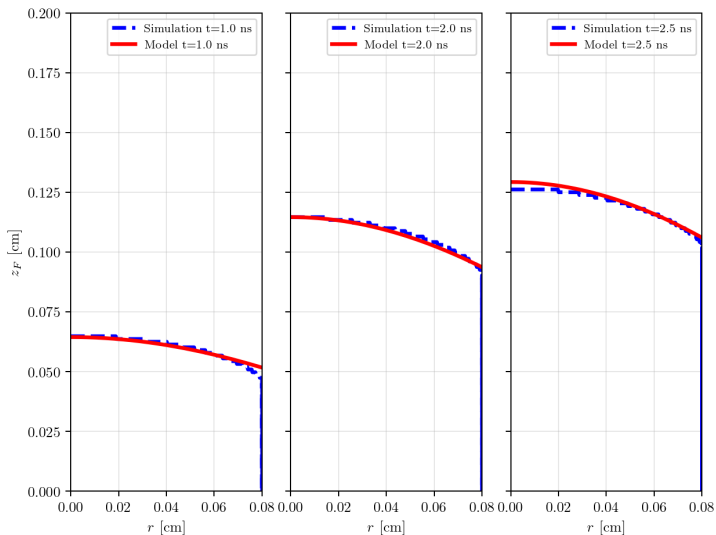
# Full 2D model

Comparing to 2D supersonic Diffusion Simulation

# Gold coating - SiO<sub>2</sub>

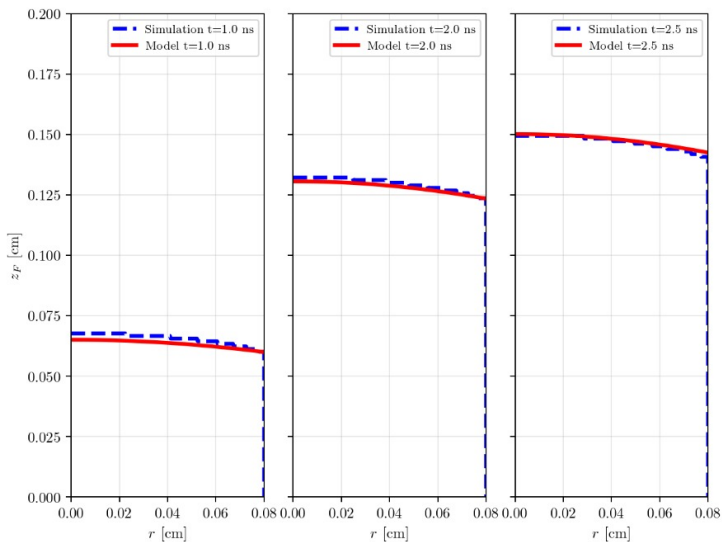


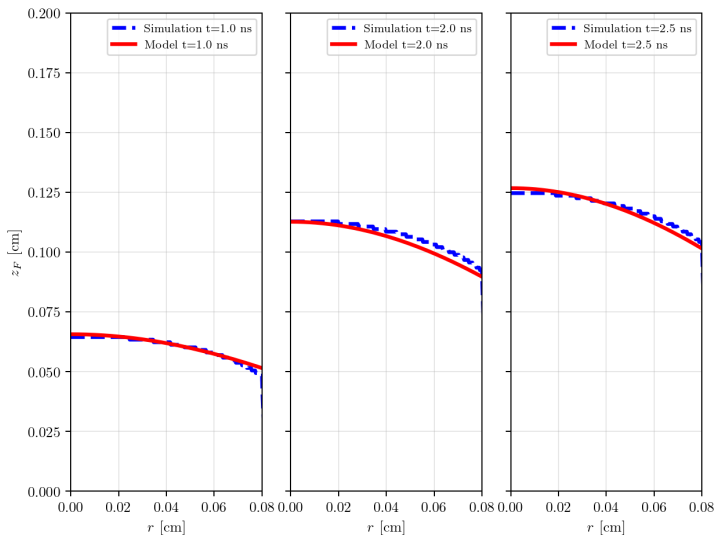
# 100 times transparenter Gold coating - SiO<sub>2</sub>



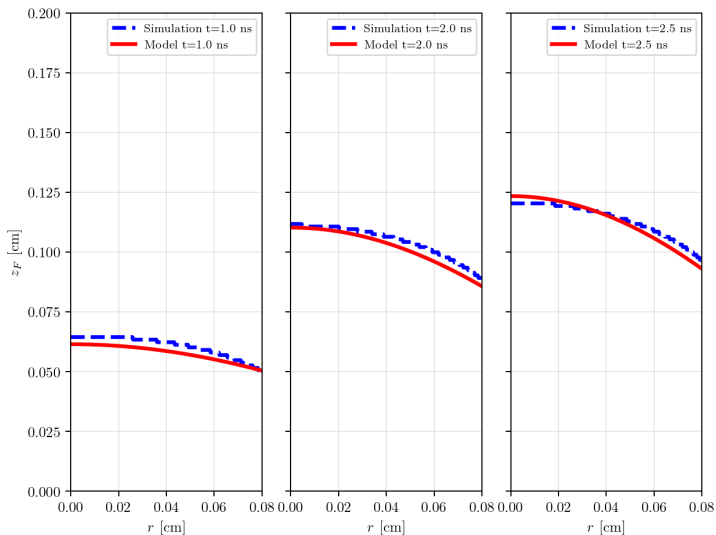


# Copper - SiO<sub>2</sub>





# Vacuum - SiO<sub>2</sub>



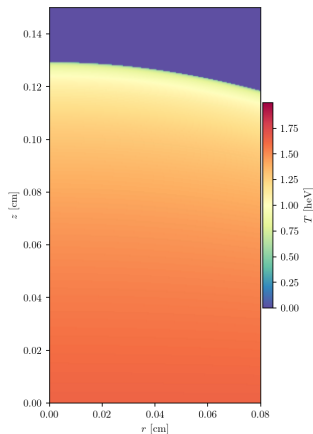
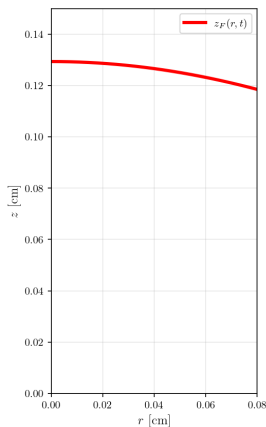
# Full 2D model

**A full 2D Temperature distribution**

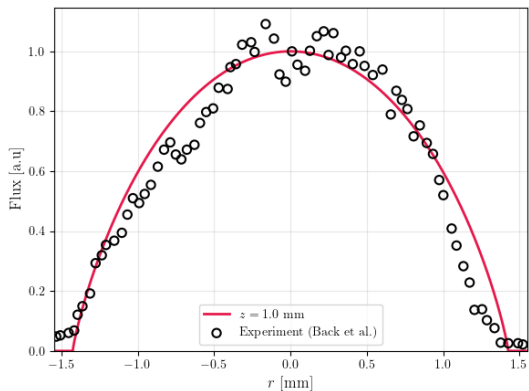
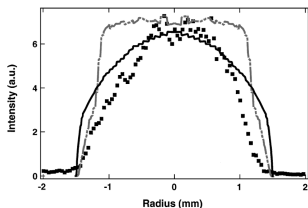
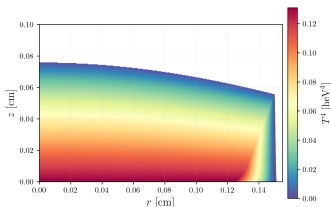
# A full 2D Temperature distribution

One can use the Heyney profile:

$$T(z, r, t) \approx T_S(t) \left[ 1 - \frac{z}{z_F(r, t)} \right]^{\frac{1}{4+\alpha-\beta}}$$

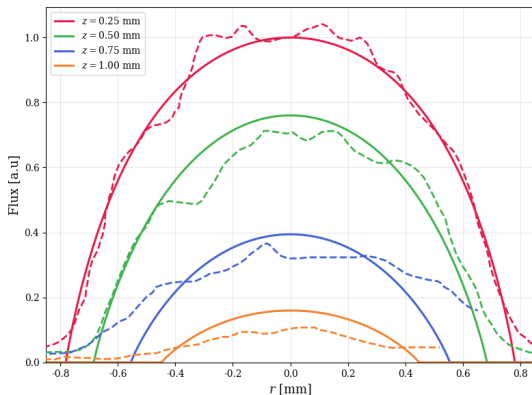
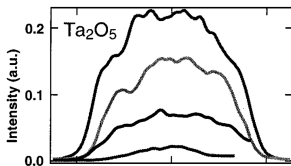
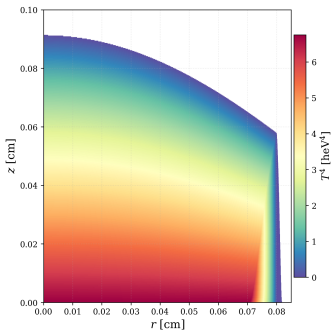


# Flux Curvature Comparison - Back SiO<sub>2</sub> PRL



*Back et al. POP, (2000)*

# Flux Curvature Comparison - Back Ta2O5 POP



*Back at el. POP, (2000)*

- Summary of key findings
- Future work

**Thank You!**

Questions?